The NIH Stage Model for Behavioral Interventions in Aging: Development, Maturation, Impact and Opportunities for Advancement

Ana-Maria Vranceanu[1,2], Ian M. Kronish[3], Evan Plys[1,2], Maggie Syme[1,2], Fang Yu[4], and Joseph E. Gaugler[5]

[1] Center for Health Outcomes and Interdisciplinary Research (CHOIR), Department of Psychiatry, Massachusetts General Hospital, Boston, MA, USA

[2] Harvard Medical School, Boston, MA, USA

[3] Department of Medicine, Columbia University

[4] Arizona State University Edson College of Nursing and Health Innovation

[5] University of Minnesota School of Public Health

Correspondence: Ana-Maria Vranceanu, PhD, Center for Health Outcomes and Interdisciplinary Research (CHOIR), Department of Psychiatry, Mass General Hospital/Harvard Medical School. One Bowdoin Square, Suite 100, Boston, MA 02114 Tel 617-724-4977; E-mail: avranceanu@mgh.harvard.edu

RUNNING HEAD: The NIH Stage Model

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Abstract:

The National Institute of Health (NIH) Stage Model provides a systematic framework for advancing behavioral intervention science. It uniquely bridges basic research with the development and dissemination of interventions that are scientifically rigorous and scalable. This manuscript examines the evolution and key features of the NIH Stage Model, its incorporation of iterative and mechanism-focused methodologies, and its impact on behavioral science, particularly in aging research. We provide examples of how the model's flexibility, iterative design, focus on mechanisms of behavior change, and emphasis on scalability are significantly influencing intervention development in aging. We describe how the National Institute of Aging (NIA) Edward R. Roybal Centers for Translational Research in the Behavioral and Social Sciences of Aging have become engines of intervention development and testing according to the NIH Stage Model by funding behavioral trials across the NIH Stage Model while simultaneously supporting the development of training opportunities and scientific networks immersed in this approach. We conclude with a discussion of strategies, fully aligned and compatible with the NIH Stage Model, to further achieve the model's ultimate goal: the swift translation of interventions, including aging interventions, into clinical, community, home, or other "real-world" settings to benefit those that need them.

Behavioral interventions have historically lacked a systematic and rigorous framework for development and often failed to address the underlying mechanisms of behavior change. This gap has hindered interventions' optimization, scalability, and real-world applicability, leading to a science-to-practice gap. The National Institute of Health (NIH) Stage Model was thus introduced, offering a translational roadmap for behavioral interventions.

The Development and Maturation of the NIH Stage Model

In the early 2000s, NIH behavioral scientists identified gaps in translational frameworks for behavioral interventions. While frameworks for pharmacological development existed, none were tailored to behavioral interventions. Rounsaville et al. (2001) proposed the "Stage Model," comprising of Stages I (intervention creation and adaptation), II (efficacy testing), and III (effectiveness testing). Around this same period, there was a growing recognition that while many efficacious and effective interventions were being developed, few of these evidence-based behavioral intervention were being implemented in communities or clinics – what is known as the science-to-service gap. To address this, the "Stage Model" was later updated and relabeled as the "NIH Stage Model". The underlying premise of the NIH Stage Model is that implementation considerations need to be integrated into behavioral intervention development. Intervention development is not complete until an intervention reaches its highest level of potency and can be implemented for the greatest number of individuals with the potential to benefit from that specific intervention (Onken et al., 2014).

Unlike frameworks like the FDA's drug development model, the NIH Stage Model is iterative, encouraging movement between stages to refine interventions based on emerging findings (Onken et al., 2014; Onken, 2019). It integrates mechanistic studies to enhance understanding of behavioral change processes and improve intervention potency (Onken et al., 2014; Onken et al 2015). Unlike the ORBIT model (Czajkowski et al., 2015), the NIH Stage Model also explicitly integrates implementation alongside

intervention development. Onken (2022) highlighted how the NIH Stage Model aligns with implementation science, emphasizing early and frequent attention to fidelity and implementability and linking mechanisms of action to practical application. The model frames Stages I–IV as focusing on developing interventions that balance potency and real-world adaptability, while Stage V emphasizes dissemination and implementation. Importantly, while each Stage includes clear activities and deliverables, the model is also non-prescriptive. Intervention developers are encouraged to work on addressing gaps in the potency and scalability based on whichever stage of development is most suitable. By linking basic science with efficacy, effectiveness, and implementation studies, the NIH Stage Model fosters cross-disciplinary collaboration and innovation.

The NIH Stage Model expects that behavioral interventions are developed with a goal of understanding of the mechanisms of behavior change. Consistent with the Experimental Medicine Approach promoted by the NIH'S Science of Behavior Change (SOBC) initiative, this involves testing mechanisms of behavior change (e.g., cognitive, affective, interpersonal, environmental) that, if influenced by the intervention components, are expected to alter the behavioral outcomes of interest. These mechanisms are hypothesized using theory and empirical evidence. Once identified, they are measured using reliable tools to test their responsiveness to experimental manipulation. Researchers then evaluate whether manipulating these mechanisms leads to desired behavioral changes, confirming their causal role. This mechanism-focused strategy enhances intervention potency early in development and supports principle-driven, scalable solutions. The NIH Stage Model also urges scientists to consider mechanisms of behavior change across all stages. This can help to maximize fidelity, as intervention potency often weakens as interventions move from efficacy to implementation testing without a dedicated focus on maintaining engagement of target mechanisms.

While the NIH Stage Model is non-linear and non-prescriptive, it still provides scientists with a roadmap from Stage 0 (basic research) to Stage V (dissemination and implementation). It does this by encouraging scientists to think carefully about what the next best Stage of intervention development will be to achieve the ultimate goal of maximally potent and scalable behavioral interventions, defined by their principles. Indeed, two groups of scientists may simultaneously be working at different Stages of development, with one group working to implement an effective intervention while another seeking to further enhance the potency or efficiency of an intervention previously shown to be effective.

The Impact of the NIH Stage Model on Aging Research

The NIH Stage Model is increasingly impactful in aging research. The NIA has actively incorporated the NIH Stage Model as a guiding framework to improve intervention development for aging populations. Training programs, such as the Summer Institute on Randomized Behavioral Trials and the Bridging the Science-to-Service Gap in Aging Care T32, use the NIH Stage Model to teach intervention development. In addition to a dementia care-relevant RFA for the Roybal Centers (RFA-AG-24-007), recent funding opportunities (e.g., PAR-21-307, PAR-21-308) and National Advisory Council on Aging (NACA)-Approved Concepts (Dementia Care and Caregiver Support Intervention Research — Stage I Only and Dementia Care and Caregiver Support Intervention Research — Stages II through V) have supported scalable dementia care interventions, addressing the needs of individuals with Alzheimer's disease and their caregivers. Across these research funding opportunities, investigators must situate their proposed interventions within a stage or stages of the NIH Stage Model and are encouraged to specify how the proposed intervention is positioned to progress to later stages of NIH Stage Model testing. In this manner, the NIH Stage Model has exerted a significant scientific impact on the rigor and implementation potential of NIA-funded behavioral intervention studies.

The NIA funded Roybal Centers clearly exemplify how the NIH Stage Model accelerates the translation of behavioral research into impactful applications for aging populations. Named after Congressman Edward R. Roybal, a co-founder of the House Select Committee on Aging and an advocate for older adult well-being (Roybal, 1988), the centers were established by Congress in 1993. Initially comprising five centers with a $3 million commitment in 2003 (National Institute on Aging, 2002), the program has expanded significantly to 15 centers with $17.25 million committed in 2024 (National Institute on Aging, n.d.). Alongside this growth, the Roybal Centers have broadened their focus to include caregiving, dementia care, and other critical aspects of aging science. The Roybal Centers have evolved in tandem with advancements in translational research practices. A notable example is the close alignment with the NIH Common Fund's Science of Behavior Change (SOBC) program, which emphasized the identification, measurement, and targeting of mechanisms of behavior change (Riddle & Science of Behavior Change Working Group, 2015). These collaborations have enhanced the capacity of Roybal Centers to ground their interventions in robust, empirically validated mechanisms of change.

In 2019, the Roybal Center program underwent a significant reorganization with the establishment of the Northwell Roybal Coordinating Center. This reorganization refined the centers' objectives to emphasize the: 1) development of principled behavioral interventions informed by basic science; 2) advancement of interventions according to the NIH Stage Model; 3) collaboration between Roybal Centers and other external centers, 4) a separate announcement for dementia care and caregiving, and 5) recruitment and mentorship of new scholars into translational research (Butler et al., 2024). These foci have catalyzed a marked increase in research activity. Since 2019, 142 publications have been generated by Roybal-funded behavioral trials and 37 grants awarded to Roybal investigators (Northwell Royball, n.d.; see Figure 1).

The Roybal Centers are designed to catalyze the translation of basic behavioral and social research in aging into real-world applications aligned with the NIH Stage Model by having annual national competitions for funding behavioral trials relevant to the theme of the Center. The initial competition after the Roybal restructuring primarily resulted in funding pilot trials for early-stage interventions. Most recently, Roybal Centers are required to fund fully powered trials. The current Roybal Centers also provide extensive training and support to enhance translational behavioral scholarship through experiential learning for awardees and the broader research community. Today, the Roybal Centers are pivotal as incubators for innovative translational behavioral science in aging. By promoting high-quality training, fostering collaboration, and providing critical funding, they continue to shape the field of aging research, bridging the gap between basic science and real-world impact.

The NIH Stage model's ongoing and future contributions to the swift translation of behavioral aging research.

Since its inception, the NIH model has been malleable and responsive to new and relevant scientific discoveries. To further increase the impact of the NIH Stage Model, we present several challenges in behavioral research in gerontology and aging that the NIH Stage Model seeks to address.

1. **Avoiding the "Implementation Cliff" through attention to dissemination and sustainability**

The NIH Stage Model emphasizes consideration of mechanisms of action and intervention delivery early in development (Onken, 2019, 2022), which has helped produce scientifically sound behavioral interventions with strong fidelity. However, despite these steps designed to increase real-world implementation, behavioral interventions often remain confined to academic centers or resource-rich clinical settings. This persistent "implementation cliff" highlights the need for guidance to ensure broader dissemination and sustainability, while upholding the non-prescriptive nature of the model.

Investigators looking for guidance on how to ensure that their interventions are implementable may want to refer to the push-pull-capacity model, which provides a framework for translating science into practice. The model identifies three conditions for successful implementation: 1) Push: A strong evidence base for the intervention and its implementation; 2) Pull: Demand for the intervention from organizations and target populations; and 3) Capacity: The ability of systems to deliver the intervention widely (Curry, 2000). Considering push, pull and capacity can be included in any Stage of the NIH Stage Model to increase attention toward dissemination and sustainability. For example, during Stages 0 and 1A, investigators could assess the need and demand for interventions in target populations and settings (i.e., establishing "pull"). This may include evaluating organizational readiness and patient interest to enhance the relevance and adoption of interventions. These early stage activities should consider challenges specific to aging research, such as workforce shortages in behavioral health and geriatric care (Buehler et al., 2024) and alignment with Center for Medicare and Medicaid Services (CMS) policy such Medicare reimbursement for services provided to persons with dementia and/or their caregivers (Lees Haggerty et al., 2020; Reuben et al., 2023).

Additionally, the engagement of non-research partners is encouraged throughout intervention development. Using participatory co-design methods in the early stages of intervention development (Moll et al., 2020) and gathering feedback from end users and decision-makers, such as patients and policymakers, can increase translatability. Ongoing engagement across NIH stages can help identify implementation barriers, inform refinements, and ensure smoother adoption without compromising mechanisms of action.

2. Avoiding an "Impact Cliff" through representation and reach

The ultimate goal of intervention research is to positively impact health and health equity. Investigators must maintain a focus on impact as the central mission of intervention development across all stages of

the NIH Stage Model. One approach to ensure this is to establish guidelines for measuring and reporting outcomes relevant to impact, including sample representativeness; clinical and community health outcomes; health equity outcomes; and reach, adoption, and sustainment (Kwan et al., 2022),

Non-representative samples can create cascading consequences throughout intervention development, reducing the likelihood of successful implementation, dissemination, sustainment, and overall impact. Prior research indicates that behavioral trials often over-sample older adults with higher education, income, and social capital (Martinson et al., 2010). Additionally, behavioral interventions may yield weaker effects for older adults with medically complex profiles and cognitive limitations (Pinquart et al., 2007). Addressing these biopsychosocial confounding factors is essential in the model's early stages (e.g., developing conceptual models that incorporate moderating factors) and later stages (e.g., subgroup analyses in efficacy trials). A strategy to counter the cascading effects of non-representative samples is prioritizing end users at the highest risk for adverse treatment outcomes in Stages 0 and I (Moll et al., 2020). For instance, in a Stage I feasibility trial, investigators should demonstrate intervention feasibility among high-risk or hard-to-reach end users (e.g., individuals with cognitive impairments, comorbidities, health-related social needs, or diverse racial and ethnic backgrounds) before advancing to Stages II or III. Such an approach could accelerate progress toward dissemination and implementation by reducing the need for Stage I revisits to develop cultural or context-sensitive adaptations.

Another critical challenge to implementation, dissemination, sustainability, and impact is the behavioral health workforce shortage, particularly among clinicians serving older adults (Buehler et al., 2024). Thus, realistically, intervention materials must be designed to be delivered by non-specialists. Just as participant representativeness must be addressed early in the development process, similar attention should be given to the representativeness/capacity/literacy of potential interventionists (e.g., professional

status, levels of geriatrics training) and care contexts (e.g., variably-resourced primary care clinics, residential long-term care settings, home and community-based programs). This aligns with the concept of designing for dissemination (Kwan et al., 2022), which may help to reduce the need to return to Stage I to revise training materials prior to a Stage III, IV, or V trial.

3. Avoiding the "Multicomponent Intervention Mechanism Challenge" by understanding the efficacy and need of each component within the interventions before efficacy testing.

Many behavioral interventions in aging are inherently multicomponent, comprising various elements designed to address complex needs. For example, interventions to improve activities of daily living in older adults often include occupational therapy, physical therapy, exercise, memory training, and cognitive-behavioral therapy (Kim et al., 2022). Fall prevention programs integrate exercise, home safety modifications, vision correction, and medication management to reduce fall risk (Sherrington et al., 2020).

Although multicomponent interventions may prove efficacious, the contributions of individual components often lack clarity. This gap hinders efforts to optimize and scale interventions, as including unnecessary or redundant components increases costs, reduces feasibility, and complicates fidelity assessments and implementation (Collins, 2018; Onken et al., 2014). Understanding how components interact—whether synergistically or antagonistically—is critical to refining multicomponent interventions. Synergistic effects occur when the combined impact of components exceeds the sum of their individual effects (Collins et al., 2009). For example, a systematic review found that combining exercise with nutritional supplements improved physical frailty in community-dwelling older adults (Sirikul et al., 2024). Conversely, some components may interact antagonistically, diminishing the overall effectiveness. A systematic review revealed that multicomponent interventions combining physical

exercises and cognitive training were more effective than either alone for individuals with normal cognition but not for those with mild cognitive impairment (Bruderer-Hofstetter et al., 2018).

With its focus on optimizing interventions for potency and scalability, the NIH Stage Model encourages testing the effects of individual components and of interactions among intervention components. Within the NIH Stage Model, investigators could consider systematic approach for evaluating and optimizing components, when appropriate, such as the Multiphase Optimization Strategy (MOST) and Sequential Multiple Assignment Randomized Trials (SMART**)** (Collins, 2018; Mohr et al., 2013). MOST employs theory-driven experimental designs, like factorial experiments, to assess the effectiveness and interactions of individual components, thereby reducing redundancy and enhancing cost-effectiveness (Collins, 2018). SMART designs facilitate the development of adaptive interventions by tailoring the combinations of components to individual needs over time (Almirall et al., 2014). For example, SMART designs is used in the FIT-AD SMART trial (clinicaltrials.gov NCT05877196), a Stage IB trial, to evaluate aerobic-fitness responses to 3-month moderate-intensity continuous training (cycling) in older adults with early Alzheimer's disease to identify non-responses. After three months, aerobic-fitness non-responders are re-randomized to high-intensity interval training or combined aerobic and resistance exercises. These systematic approaches increase the likelihood that interventions advancing to later stages will be both effective and feasible for real-world application, ultimately enhancing the impact of aging-related interventions.

4. Avoiding the lag between effectiveness and implementation trials through incorporation of hybrid designs within the NIH Stage model.

The NIH Stage Model is also flexible in accommodating methodological approaches that leverage implementation-relevant data collected in earlier stages (e.g., Stages I-III) to inform an intervention's

contextual "fit" for real-world application. For example, hybrid effectiveness-implementation studies integrate assessments of effectiveness with measures of implementation success within a single trial (Curran et al., 2012, 2022). By simultaneously addressing internal validity (e.g., efficacy and effectiveness outcomes) and external validity (e.g., feasibility, acceptability, fidelity, and appropriateness for partner organizations), hybrid studies generate holistic evidence that expedites intervention development and dissemination, improving efficiency across the translational pipeline (Curran et al., 2012, 2022).

Hybrid effectiveness-implementation studies are categorized into three types (I-III) based on the prioritization of implementation. Type I trials prioritize intervention efficacy/effectiveness while collecting implementation-relevant data (e.g., feasibility and appropriateness for organizations) through mixed methods designs (Creswell & Clark, 2017); Type II trials balance evaluation of both intervention efficacy/effectiveness and implementation strategies, often using designs like a 2x2 factorial experiment to measure outcomes across intervention/usual care and implementation strategy/no strategy groups. Type III trials primarily test implementation strategies' effectiveness (e.g., adoption, fidelity, or sustainability) on implementation outcomes.

Hybrid effectiveness are valuable across stages of the NIH Stage Model, as long as the scientific justification for these studies are appropriate. For instance, in a Stage I dementia care pilot trial, Gaugler et al. (2024) are utilizing a Type I hybrid design to assess: a) the intervention's impact (fully powered) on mechanisms of action and well-being outcomes for people with cognitive impairment living alone; and b) implementation-relevant domains such as acceptability and feasibility through embedded qualitative and quantitative data collection. This approach facilitates iterative adaptation and positions the intervention for optimal dissemination and implementation in later stages (University of Minnesota, 2023). Similarly, Vranceanu and colleagues (McDermott et al., 2023) conducted an R61/R33 trial that culturally tailored a mind-body activity program for older adults in an underserved community clinic. The R61 phase pilot-

tested a culturally tailored version of an efficacious intervention while collecting qualitative data from providers and participants using implementation frameworks. In the R33 phase, the team transitioned to a fully powered effectiveness-implementation Type I trial to evaluate the program's effects on physical function and implementation outcomes.

5. Avoiding the tension between fidelity, adaptation, and progression through the NIH Stage Model

A key challenge in advancing aging interventions through the NIH Stage Model is balancing intervention fidelity, essential for internal validity, with adaptation to fit partner organizations' resources, expertise, and cultural relevance during Stage IV and V testing. Limited understanding of "essential ingredients"—the elements responsible for demonstrated benefits in earlier trials—hinders progress. While adaptation is necessary to align interventions with an organization's context, altering essential ingredients risks compromising their benefits.

Early-stage intervention development should focus on mapping an intervention's "forms and function." Function refers to how intervention components influence outcomes, while forms are the specific activities of delivery (who, what, when, where, and how). Differentiating form and function is often overlooked in aging interventions. If this understanding is developed early through theory and testing, later adaptations can enhance fit within diverse contexts while preserving essential elements. This deliberate approach ensures implementation success without sacrificing effectiveness.

Research on Patient-Centered Medical Homes (PCMHs) illustrates this method. Previous studies showed inconsistent benefits due to the model's complexity and unclear mechanisms of action (Perez Jolles et al., 2019). To address this, Perez Jolles et al. (2019) conducted a scoping review to identify hypothesized core mechanisms and developed the PCMH Function and Form Matrix, a theoretical framework outlining how PCMHs achieve their effects. Using such tools earlier in the NIH Stage Model

allows researchers to separate essential elements from adaptable features, enhancing scalability and successful implementation. This strategy ensures interventions remain effective while adapting to diverse contexts.

6. Avoiding delays due to "Funding mechanisms challenges"

The process of intervention development, testing, and successful dissemination/implementation averages 17 years (Munro & Savel, 2016). Part of this delay in a US context is attributable to the structure of current federal funding mechanisms. For instance, Stage I research, which focuses on initial intervention development and pilot testing, is often supported by R21 grants. These grants provide limited resources, which can constrain researchers' ability to conduct rigorous studies. If an intervention's feasibility is found to be low, additional pilot studies may be required. Similarly, if underlying mechanisms fail to support the intervention, researchers may need to return to Stage 0 for further exploration. Both scenarios demand significant time and resources, often exceeding the capacity of typical R21 funding.

A critical barrier to progressing through the NIH Stage Model is the limited funding mechanisms that support seamless advancement between stages (e.g., from Stage I to Stage V). Most funding mechanisms are designed to support only one stage at a time, requiring researchers to submit separate grant applications for each subsequent trial. This piecemeal approach introduces delays, disrupts research momentum, and hinders the development of cohesive research programs. While phased grant mechanisms such as R21/R33 and R61/R33 provide a bridge between initial pilot work and more substantial research, their availability is limited. Expanding these mechanisms could help researchers navigate the Stage Model more effectively and avoid prolonged delays.

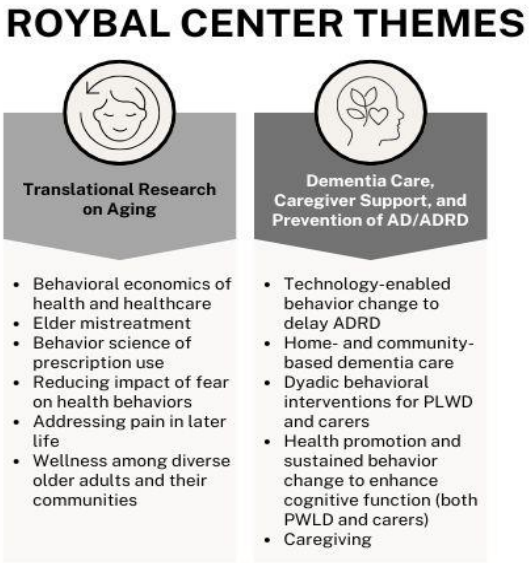
To address these challenges, developing additional phased Program Announcements with Special Review (PARs) and Requests for Applications (RFAs) for all stages of the NIH Stage Model (I through V) is essential. These funding opportunities would provide researchers with the continuity and resources

needed to efficiently advance behavioral interventions across the stages, ensuring that promising interventions progress to implementation and dissemination without the typical 1–2-year delays associated with securing new funding after a grant ends. This streamlined approach would enhance behavioral/aging intervention research's efficiency, innovation, and rigor.

Summary and Conclusions

The NIH Stage Model provides a systematic framework for developing, testing, and disseminating behavioral interventions, particularly in aging research. The NIH Stage Model's emphasis on iterative movement across stages ensures that interventions are rigorously tested for feasibility, fidelity, efficacy, and scalability. The model has served as a transformative milestone in the science of behavioral interventions in aging (and NIA's 50th anniversary) through its accommodation of multiple methodological innovations to further facilitate interventions' progression across its stages (e.g., MOST and SMART designs; identification and testing of mechanisms to facilitate subsequent implementation efforts, hybrid effectiveness studies). By emphasizing principles, the NIH Stage Model bridges gaps between intervention efforts across fields. Although more general challenges in the design, testing, implementation, and funding of interventions for older people and those who care for them remain, the NIH Stage Model has propelled the science of behavioral interventions in aging forward through its support of rigorous and dynamic design and the strategic development of interventions throughout their developmental lifecycle so that scientifically-based interventions may realize their full public health potential.

Figure 1. The themes of the active Roybal Centers



References:

Almirall, D., Nahum-Shani, I., Sherwood, N. E., & Murphy, S. A. (2014). Introduction to SMART designs for the development of adaptive interventions: With application to weight loss research. *Translational Behavioral Medicine*, *4*(3), 260–274. https://doi.org/10.1007/s13142-014-0265-0

Bruderer-Hofstetter, M., Rausch-Osthoff, A.-K., Meichtry, A., Münzer, T., & Niedermann, K. (2018). Effective multicomponent interventions in comparison to active control and no interventions on physical capacity, cognitive function and instrumental activities of daily living in elderly people with and without mild impaired cognition—A systematic review and network meta-analysis. *Ageing Research Reviews*, *45*, 1–14. https://doi.org/10.1016/j.arr.2018.04.002

Buehler, S., Emery-Tiburcio, E. E., Buehler, S., & Emery-Tiburcio, E. E. (2024). Building the Geriatric Mental Health and Substance Use Workforce. *Generations*, *1*, Article 1.

Butler, M. J., Davidson, K. W., & Suls, J. (2024). The National Institute on Aging Edward R. Roybal Centers: Past, present, and future of research to foster healthy aging. *Journal of the American Geriatrics Society*, *72*(5), 1590–1594. https://doi.org/10.1111/jgs.18865

Collins, L. M. (2018). *Optimization of Behavioral, Biobehavioral, and Biomedical Interventions: The Multiphase Optimization Strategy (MOST)*. Springer International Publishing. https://doi.org/10.1007/978-3-319-72206-1

Collins, L. M., Dziak, J. J., & Li, R. (2009). Design of experiments with multiple independent variables: A resource management perspective on complete and reduced factorial designs. *Psychological Methods*, *14*(3), 202–224. https://doi.org/10.1037/a0015826

Creswell, J. W., & Clark, V. L. P. (2017). *Designing and Conducting Mixed Methods Research* (3rd ed.). Sage Publications. https://collegepublishing.sagepub.com/products/designing-and-conducting-mixed-methods-research-3-241842

Curran, G. M., Bauer, M., Mittman, B., Pyne, J. M., & Stetler, C. (2012). Effectiveness-implementation hybrid designs: Combining elements of clinical effectiveness and implementation research to enhance public health impact. *Medical Care*, *50*(3), 217–226. https://doi.org/10.1097/MLR.0b013e3182408812

Curran, G. M., Landes, S. J., McBain, S. A., Pyne, J. M., Smith, J. D., Fernandez, M. E., Chambers, D. A., & Mittman, B. S. (2022). Reflections on 10 years of effectiveness-implementation hybrid studies. *Frontiers in Health Services*, *2*. https://doi.org/10.3389/frhs.2022.1053496

Curry, S. J. (2000). Organizational interventions to encourage guideline implementation. *Chest*, *118*(2 Suppl), 40S-46S. https://doi.org/10.1378/chest.118.2_suppl.40s

Czajkowski, S. M., Powell, L. H., Adler, N., Naar-King, S., Reynolds, K. D., Hunter, C. M., Laraia, B., Olster, D. H., Perna, F. M., Peterson, J. C., Epel, E., Boyington, J. E., & Charlson, M. E. (2015). From ideas to efficacy: The ORBIT model for developing behavioral treatments for chronic diseases. *Health Psychology: Official Journal of the Division of Health Psychology, American Psychological Association*, *34*(10), 971–982. https://doi.org/10.1037/hea0000161

Gaugler, J. E., Baier, R. R., Baker, Z. G., Boltz, M., Fortinsky, R. H., Gustavson, A. M., Hodgson, N. A., Jutkowitz, E., McPhillips, M. V., Parker, L. J., Sefcik, J. S., & Gitlin, L. N. (2024). Using Hybrid Effectiveness Studies to Facilitate Implementation in Community-Based Settings: Three Case Studies in Dementia Care Research. *Journal of the American Medical Directors Association*, *25*(1), 27–33. https://doi.org/10.1016/j.jamda.2023.07.025

Kim, M., Shin, E., Kim, S., & Sok, S. (2022). The Effectiveness of Multicomponent Intervention on Daily Functioning among the Community-Dwelling Elderly: A Systematic Review. *International Journal of Environmental Research and Public Health*, *19*(12), 7483. https://doi.org/10.3390/ijerph19127483

Kwan, B. M., Brownson, R. C., Glasgow, R. E., Morrato, E. H., & Luke, D. A. (2022). Designing for Dissemination and Sustainability to Promote Equitable Impacts on Health. *Annual Review of Public Health*, *43*, 331–353. https://doi.org/10.1146/annurev-publhealth-052220-112457

Lees Haggerty, K., Epstein-Lubow, G., Spragens, L. H., Stoeckle, R. J., Evertson, L. C., Jennings, L. A., & Reuben, D. B. (2020). Recommendations to Improve Payment Policies for Comprehensive Dementia Care. *Journal of the American Geriatrics Society*, *68*(11), 2478–2485. https://doi.org/10.1111/jgs.16807

Martinson, B. C., Crain, A. L., Sherwood, N. E., Hayes, M., Pronk, N. P., & O'Connor, P. J. (2010). Population Reach and Recruitment Bias in a Maintenance RCT in Physically Active Older Adults. *Journal of Physical Activity & Health*, *7*(1), 127–135.

McDermott, K., Levey, N., Brewer, J., Ehmann, M., Hooker, J. E., Pasinski, R., Yousif, N., Raju, V., Gholston, M., Greenberg, J., Ritchie, C. S., & Vranceanu, A.-M. (2023). Improving Health for Older Adults With Pain Through Engagement: Protocol for Tailoring and Open Pilot Testing of a Mind-Body Activity Program Delivered Within Shared Medical Visits in an Underserved Community Clinic. *JMIR Research Protocols*, *12*, e52117. https://doi.org/10.2196/52117

Mohr, D. C., Cheung, K., Schueller, S. M., Hendricks Brown, C., & Duan, N. (2013). Continuous evaluation of evolving behavioral intervention technologies. *American Journal of Preventive Medicine*, *45*(4), 517–523. https://doi.org/10.1016/j.amepre.2013.06.006

Moll, S., Wyndham-West, M., Mulvale, G., Park, S., Buettgen, A., Phoenix, M., Fleisig, R., & Bruce, E. (2020). Are you really doing 'codesign'? Critical reflections when working with vulnerable populations. *BMJ Open*, *10*(11), e038339. https://doi.org/10.1136/bmjopen-2020-038339

Munro, C. L., & Savel, R. H. (2016). Narrowing the 17-Year Research to Practice Gap. *American Journal of Critical Care: An Official Publication, American Association of Critical-Care Nurses*, *25*(3), 194–196. https://doi.org/10.4037/ajcc2016449

National Institute on Aging. (n.d.). *Frequently Asked Questions for the Edward R. Roybal Centers for Translational Research in the Behavioral and Social Sciences of Aging*. National Institute on Aging. Retrieved December 17, 2024, from https://www.nia.nih.gov/research/dbsr/frequently-asked-questions-edward-r-roybal-centers-translational-research-behavioral

National Institute on Aging. (2002, October 28). *EDWARD R. ROYBAL CENTERS FOR TRANSLATIONAL RESEARCH IN THE BEHAVIORAL AND SOCIAL SCIENCES*. https://grants.nih.gov/grants/guide/rfa-files/RFA-AG-03-002.html

Northwell Royball. (n.d.). *Northwell Roybal Coordinating Center to Accelerate Translational Aging Research*. Northwell Roybal About Us. Retrieved December 17, 2024, from https://www.roybalniaresearchcenters.org/about-us

Onken, L. (2019). History and evolution of the NIH stage model: Overcoming hurdles to create behavioral interventions to improve the public health. In *Evidence-Based Practice in Action Bridging Clinical Science and Intervention* (pp. 28–43). The Guilford Press. https://www.guilford.com/books/Evidence-Based-Practice-in-Action/Sona-Dimidjian/9781462547708/contents

Onken, L. (2022). Implementation Science at the National Institute on Aging: The Principles of It. *Public Policy & Aging Report*, *32*(1), 39–41. https://doi.org/10.1093/ppar/prab034

Onken, L., Carroll, K., Shoham, V., Cuthbert, B., & Riddle, M. (2014). Reenvisioning Clinical Science: Unifying the Discipline to Improve the Public Health. *Clinical Psychological Science : A Journal of the Association for Psychological Science*, *1*(2), 22–34.

Onken, L. S. (2015). Cognitive Training: Targeting Cognitive Processes in the Development of Behavioral Interventions. *Clinical Psychological Science*, *3*(1), 39–44. https://doi.org/10.1177/2167702614561512

Perez Jolles, M., Lengnick-Hall, R., & Mittman, B. S. (2019). Core Functions and Forms of Complex Health Interventions: A Patient-Centered Medical Home Illustration. *Journal of General Internal Medicine*, *34*(6), 1032–1038. https://doi.org/10.1007/s11606-018-4818-7

Pinquart, M., Duberstein, P. R., & Lyness, J. M. (2007). Effects of psychotherapy and other behavioral interventions on clinically depressed older adults: A meta-analysis. *Aging & Mental Health*, *11*(6), 645–657. https://doi.org/10.1080/13607860701529635

Reuben, D. B., Epstein-Lubow, G., & Super, N. (2023). The Other Dementia Breakthrough-Comprehensive Dementia Care. *JAMA Neurology*, *80*(8), 770–772. https://doi.org/10.1001/jamaneurol.2023.1252

Riddle, M. & Science of Behavior Change Working Group. (2015). News from the NIH: Using an experimental medicine approach to facilitate translational research. *Translational Behavioral Medicine*, *5*(4), 486–488. https://doi.org/10.1007/s13142-015-0333-0

Rounsaville, B. J., Carroll, K. M., & Onken, L. S. (2001). A stage model of behavioral therapies research: Getting started and moving on from stage I. *Clinical Psychology: Science and Practice*, *8*(2), 133–142. https://doi.org/10.1093/clipsy.8.2.133

Roybal, E. R. (1988). Mental health and aging: The need for an expanded federal response. *American Psychologist*, *43*(3), 189–194. https://doi.org/10.1037/0003-066X.43.3.189

Sherrington, C., Fairhall, N., Kwok, W., Wallbank, G., Tiedemann, A., Michaleff, Z. A., Ng, C. A. C. M., & Bauman, A. (2020). Evidence on physical activity and falls prevention for people aged 65+ years: Systematic review to inform the WHO guidelines on physical activity and sedentary

behaviour. *International Journal of Behavioral Nutrition and Physical Activity*, *17*(1), 144. https://doi.org/10.1186/s12966-020-01041-3

Sirikul, W., Buawangpong, N., Pinyopornpanish, K., & Siviroj, P. (2024). Impact of multicomponent exercise and nutritional supplement interventions for improving physical frailty in community-dwelling older adults: A systematic review and meta-analysis. *BMC Geriatrics*, *24*(1), 958. https://doi.org/10.1186/s12877-024-05551-8

University of Minnesota. (2023). *School of Public Health University of Minnesota*. School of Public Health - Families and Long-Term Care. https://www.sph.umn.edu/research/projects/ltc/